Practice Sheet – 1

Probability and Statistics

BMAS 0108

1. Which average would be suitable in the following cases?
   (i) Average size of readymade garments.
   (ii) Average intelligence of students in a class.
   (iii) Average production in a factory per shift.
   (iv) Average wages in an industrial concern.
   (v) When the sum of absolute deviations from average is least.
   (vi) When quantities of the variable are in ratios.
   (vii) In case of open-ended frequency distribution.

2. Given below is the distribution of marks obtained by 140 students in an examination:

Marks	:	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of Students	:	7	15	18	25	30	20	16	7	2

   Find the mean and Median of the distribution.

3. For the two frequency distributions given below, the mean calculated from the first was 25.4 and that from the second was 32.5. Find the values of x and y.

Class	Distribution I	Distribution II
10-20	20	4
20-30	15	8
30-40	10	4
40-50	x	2x
50-60	y	y

4. The mean of the following frequency distribution is 50. But the frequencies f_1 and f_2 in classes 20-40 and 60-80 are missing. Find the missing frequencies.

Class :	0-20	20-40	40-60	60-80	80-100	Total
Frequency	17	f_1	32	f_2	19	120

5. An incomplete distribution is given below :

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Total
Frequency	12	30	?	65	?	25	18	229

   Find out missing frequencies if the median value is 46.

6. The distribution of age of patients turned out in a hospital on January 1, 2005, was as follows:

Age (in Years)	No. of Patients
More than 10	148
More than 20	124
More than 30	109
More than 40	71
More than 50	30
More than 60	16
More than 70 and up to 80	01

Find the median age and modal age of the patients.

7. For a group of 200 candidates, the mean arid standard deviation of scores were found to be 40 and 15 respectively. Later on it was discovered that the scores 43 and 35 were misread as 34 and 53 respectively. Find the corrected mean and standard deviation corresponding to the corrected figures.

8. Given below is the distribution of 140 candidates obtaining marks X or higher in it certain examinations (all marks are given in whole numbers) :

   X: 10 20 30 40 50 60 70 80 90 100

   c.f. : 140 133 118 100 75 45 15 9 2 0

   Calculate the mean, median, and mode of the distribution.

9. Ten percent of the workers in a firm employing a total of 1000 workers earn less than Rs. 50 per day, 200 workers earn between Rs. 50 and 99.99, thirty percent between 100 and 149.99, 250 workers between 150 and 199.99 and the rest Rs. 200 and above. Summarise this information in the form of a frequency distribution. Also find the median average.

10. In an examination of 675 candidates the examiner supplied the following information:

    Marks (Less than): 10% 20% 30% 40% 50% 60% 70% 80%
    No. of Candidates: 7 39 95 201 381 545 631 675
    Calculate the mode and median of the percentage marks obtained.

11. Calculate the quartile deviation for the following frequency distribution:
    Class intervals: 5-7 8-10 11-13 14-16 17-19
    Frequency : 14 24 38 20 4

12. Find the interquartile range and the coefficients of quartile deviation from the following data:
    Size of item: 16-20 21-25 26-30 31-35 36-40 41-45 46-50
    Frequency : 10 15 20 25 10 12 8

13. Mean and Standard deviation of the following continuous series are 31 and 15.9 respectively. The distribution after taking step deviation is as follows:

d :	-3	-2	-1	0	1	2
f :	10	15	25	25	10	10

Determine the actual class intervals.

14. The mean of 5 observations is 4.4 and the variance is 8.24. if three of the five observations are 1, 2, and 6, Find the values of the other two.

15. The following is the record of goals scored by team 'A' in a football season:

No. of goals scored :	0	1	2	3	4
No. of matched played :	1	9	7	5	3

For team 'B' the average number of goals scored per match was 2.5 with a standard deviation of 1.25 goals. Find which team may be considered more consistent?

16. Calculate the mean deviation from the mean for the following data:

Marks :	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students:	6	5	8	15	7	6	3